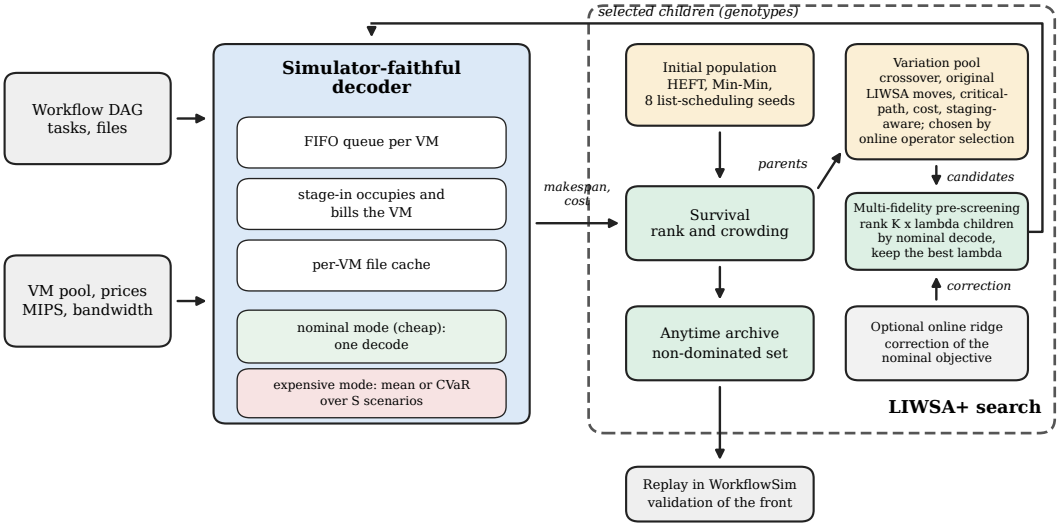
selected children (genotypes)
Workflow DAG
tasks, files
VM pool, prices
MIPS, bandwidth
Simulator-faithful decoder
FIFO queue per VM
stage-in occupies and bills the VM
per-VM file cache
nominal mode (cheap): one decode
expensive mode: mean or CVaR over S scenarios
makespan, cost
Initial population
HEFT, Min-Min,
8 list-scheduling seeds
parents
Survival
rank and crowding
Anytime archive
non-dominated set
Variation pool
crossover, original LIWSA moves, critical-path, cost, staging-aware; chosen by online operator selection
candidates
Multi-fidelity pre-screening
rank K x lambda children by nominal decode, keep the best lambda
correction
Optional online ridge correction of the nominal objective
LIWSA+ search
Replay in WorkflowSim
validation of the front